Dr. Kevin Godin-Dubois

Contact

kgd-al.github.io
k.j.m.godin-dubois@vu.nl
Vrije Universiteit Amsterdam
de Boelelaan 1081a, 1081HV
Amsterdam, The Netherlands

kgd-al@github.com
0009-0002-6033-3555
kgd-alife@LinkedIn
kgd-al

Last position Researcher in Evolutionary Robotics (until November 2026)

Highlights

Teaching 9 years (∼800 hours)

Locations: Toulouse I University & Toulouse III University, France; Vrije Universiteit Amsterdam, The Netherlands

Missions: Course design & administration, TA coordination, lectures & lab teaching, exam design & grading, project & thesis supervision ...

Computer Science:
Research thesis supervision (8 BSc, 3 MSc, full list)
Learning Machines, NeuroEvolution
Programming languages: Python, C, R
Algorithms, Data Structures, Information theory
Programming projects

Generalists:
Data Science tools and languages
Database modelling, SQL

Positions

Postdoctoral
Oct. 2026
Nov. 2022
Computer Science / Evolutionary Robotics
"NeuroEvolution and Reinforcement Learning for Embodied Robots"
Computational Intelligence Group - Vrije Universiteit Amsterdam, The Netherlands
Supervisor: Dr. A. Yaman (a.yaman@vu.nl)
Collaborators: Dr. A. Kononova (a.kononova@liacs.leidenuniv.nl)

Postdoctoral
Oct. 2022
Jul. 2020
Computer Science / Artificial Intelligence
"Emergent cognitive architectures in virtual embodied robots"
REVA Team, IRIT - Toulouse I University, France
Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr)
Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)

PhD
Jul. 2020
Sep. 2016
Computer Science / Artificial Life
"Environment-driven speciation: long term interactions in artificial plant communities"
REVA Team, IRIT - Toulouse I University, France
Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr)
Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)

Research	Artificial Life: Morphology, Cognition & Interaction
Output	3 journal articles (*Artificial Life, Frontiers, JOSS*) 7 international conference articles (*ALife, IEEE ALife, EvoAPP...*) 7 international workshops short papers (*ALife, ECAL, AAAI*) Scientific software: ABrain, AMaze, Revolve2, SHARPIE
Skills	
Programming	Fluent: C++, Bash, Python, LaTeX Working Knowledge: C, Java, R, VB, VBA, JavaScript Frameworks: (Py)Qt, MuJoCo
Technical	Evolutionary Algorithms, Machine Learning, Multi-Agents Systems, High-Performance Computing
Languages	French (native), English (professional command)

Teachings

Postdoc	Vrije Universiteit Amsterdam	
Feb. 2026	*Fully international*	
Jun. 2023	• NeuroEvolution (cohort-wide lectures)	*4h*
	• Learning Machines (projects)	*45h*
	• Master and Bachelor thesis supervision (list)	*∼200h*
Course management	Toulouse I University & Toulouse III University, France	
	• Computer Science projects	*72h*
Jul. 2022	*Multi-Agent Systems, Complex Systems, Simulation*	
Jan. 2021	• R programming (English, cohort-wide lectures)	*67.5h*
	• Information theory	*22.5h*
	• Servers and contents	*18.75h*
Teaching fellow	Toulouse I University & Toulouse III University, France	
Jul. 2020	• Statistical software (R & Python)	*36h*
Jan. 2017	• Algorithms (with project-based evaluation)	*60h*
	• Excel & VBA	*60h*
	• Modelling in databases	*21h*
Practical work supervisor	Toulouse III University, France	
	• Software projects	*69.2h*
Jun. 2021	• Data structures	*18.8h*
Sep. 2016	• C Programming	*36h*
	• Python	*8h*

Research

Research fields	• Evolutionary Robotics • Reinforcement Learning • Artificial Neural Networks • Species dynamics • Morphogenetic Engineering

Expertise	
Evolutionary Algorithms	• Environment-Driven Evolutionary Selection (EDEnS) • Multi-objective Optimisation (NSGA-II) • Vector-based Optimisation (CMA-ES) • High Performance Computing (HPC), Co-evolution, Novelty
Machine Learning	• Artificial Neural Networks (ANN, CNN, RNN) • Composite Pattern-Producing Networks (CPPN) • Cartesian Genetic Programming (CGP) • Genetic Regulatory Networks (GRN) • Hidden Markov Models (HMM) • Stable baselines 3, PyTorch
Physics simulators	• Approximate: Bullet3, Box2D • Exact: MuJoCo

Outreach

Reviewer Since 2023	Journals: • Artificial Life • IEEE Transactions on Evolutionary Computation • Journal of Open Source Software (JOSS) Conferences: • International Conference on Artificial Life (ALife) • International Conference on Intelligent Robots and Systems (IROS) • Parallel Problem Solving from Nature (PPSN)
EduMix Aspi-Friendly 2021	Initiated a project for the self-monitoring of well-being in students with autistic disorders alongside a heterogeneous team of neuro-(a)typical and various profiles (faculty, designers, developers ...).

Internships

Morphogenetic Engineering Jan. - Jul. 2016	Toulouse Research Institute on Computer Science (IRIT), France *"Rule-based artificial embryogenesis in a complex 3D environment"* Deployed rule-based genomes on the MecaCell platform to study artificial plant growth and cell specialisation. **Contact:** Pr. Y. Duthen (yves.duthen@irit.fr)
Machine Learning Apr. - Jul. 2015	IRIT, *"Comparison of different evolutionary approaches, an application to the GECCO 2015 challenge"* Performed a performance comparison (accuracy, efficiency) between Artificial Neural and Genetic Regulatory Networks on the 2015 GECCO temperature prediction challenge data. **Contact:** Pr. H. Luga (herve.luga@irit.fr)

Machine Learning May - Jul. 2014	IRIT, *"An architecture for automated bird discrimination"* Applied Hidden Markov Models to the BirdClef2014 challenge on the identification of specific bird species in a corpus of thousands of recordings. **Contact:** Pr. J. Farinas (jerome.farinas@irit.fr)

Education

PhD Jul. 2020 Sep. 2016	Toulouse I University, France Defended the 15th of July 2020 Thesis title: *"Environment-driven speciation: long term interactions in artificial plant communities"* Investigated how complexification of artificial creatures could be further enhanced through the indirect control provided by a co-evolved, highly dynamical environment. **Rapporteurs:** Pr. P. Collet & DoR. F. Vico **Contact:** Pr. Y. Duthen (yves.duthen@irit.fr)
Master 2014 - 2016	Toulouse III University, France (*with honours*) Artificial Intelligence: mathematical & symbolic models, training methods
Bachelor 2011 - 2014	Toulouse III University (*with distinction*) Computer Science: networks, programming, systems, mathematics

Scholarships and Fellowships

2023 - 2026 $\sim$ 200K €	Postdoctoral funding from the Hybrid Intelligence Centre (Netherlands)
2016 - 2019 70K €	PhD Fellowship from the French Minister of Higher Education and Research (MESR)
2015 10K €	Master Scholarship from the International Mathematics and Computer Science Center (LabEx CIMI, Toulouse)
2014 - 2015 3.6K €	Merit Scholarship from the Regional Student Welfare Office (CROUS, Toulouse)